

Beantol Messenger AI Bot

Complete Project Overview & Feature Guide

Prepared for Beantol Coffee Roasters · June 2026

Facebook Messenger + Instagram DMs · Node.js · OpenAI · Google Workspace · Render

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1. Executive Summary

The Beantol Messenger AI Bot is a **24/7 AI sales and customer support assistant** that lives inside your Facebook Page and Instagram inbox. It answers product questions, helps customers choose beans, generates quotes, books shop visits, captures leads and orders, and escalates to human staff when needed — all without making customers leave Messenger or Instagram.

This document explains **what the bot does, how it works, where to change things**, and **what was built** — useful for internal review or for showing other shop owners what a similar system could do for their business.

Platform: Facebook Messenger + Instagram DMs

Hosting: Render (beantol-bot.onrender.com)

AI: OpenAI gpt-4o-mini with RAG knowledge search

Data: Google Docs (knowledge) + Google Sheets (CRM)

2. What Problem It Solves

Before	With the bot
Staff must reply manually to every DM	AI handles FAQs, pricing, recommendations instantly
Product knowledge scattered in chats	Central knowledge base (Google Doc) synced to AI
Orders and inquiries lost in messages	Leads, orders, quotes saved to Google Sheets
No after-hours coverage	AI replies anytime; live agents 9 AM–9 PM
Hard to track who needs follow-up	Admin dashboard + email alerts + sales pipeline

3. Channels Supported

Channel	Status	Notes
Facebook Messenger	Live	Primary channel; webhook + Page token
Instagram DMs	Live	Same bot, same admin dashboard
WhatsApp	Not built	Possible future addition
Website chat widget	Not built	Possible future addition

4. How It Works



Message processing order (simplified)

1. Admin takeover — skip if staff is actively chatting
2. Get Started / first-time welcome
3. Reply-to context (swipe-reply YES on a specific bot message)
4. Agent handoff acceptance
5. Quote confirmation flow
6. Post-quote follow-up
7. Payment proof handling
8. Outside-Cebu delivery rules
9. Cebu area delivery (Maxim vs Naga vs province)
10. Weekend pickup vs delivery rules
11. Delivery rep YES acceptance
12. Phrase-based human handoff
13. Appointment booking wizard
14. Product recommendation wizard
15. **AI reply** (RAG + rules + inventory + history)
16. Post-AI quote generation if pricing was discussed

Critical business rules run **before** the AI, so the bot stays consistent even when the AI would otherwise improvise.

5. Feature List

5.1 AI Customer Support

- Answers hours, location, products, pricing, payment, brewing, storage, contacts
- Replies in English (even if customer writes Cebuano/Tagalog/Bislish)
- Conversation memory for context across messages
- RAG search of Google Doc — does not guess prices or products

5.2 AI Sales Assistant

- Consultative selling: brew method, taste, volume
- Recommends 1–2 beans with explanations
- Correct retail sizes and wholesale rules (6 kg MOQ for Prime, Santos, Cerrado only)
- Respects out-of-stock inventory
- Follow-up for stale quoted leads

5.3 Product Recommendations

Guided step-by-step picker: brew method → taste → 1–2 bean suggestions with prices.

5.4 Quotation System

- Formal quotes with ID (QT-...) and printable shareable page
- Saved to Google Sheets Quotes tab

5.5 Delivery Workflows

Zone	Method
Cebu City, Mandaue, Talisay, Lapu-Lapu	Maxim (customer pays rider fee); payment + proof required first
Remote Cebu Province (Naga, Carcar, etc.)	Pickup, own logistics, or J&T/courier
Outside Cebu / nationwide	J&T or preferred courier; team confirms cost/time

Delivery alerts email the team. Delivery does **not** pause the bot.

5.6 Human Handoff

- Triggers on phrases like "talk to a human", "real person", "need agent"
- Email alert + dashboard shows "Awaiting agent"
- AI keeps answering until staff replies in Business Suite → bot pauses
- Auto-resumes after 15 min admin idle, or manual Resume AI
- Live agents: 9 AM–9 PM Philippine time daily

5.7 Appointment Booking

- Shop visit booking via chat
- Natural dates: "today 5pm", "tomorrow", "this coming Thursday"
- Saved to Google Sheets Appointments tab

5.8 Lead, Order & Event Capture

Data	Storage	Admin
Leads	Google Sheet	/admin/leads
Orders	Google Sheet	/admin/orders
Quotes	Google Sheet	/admin/quotes
Appointments	Google Sheet	/admin/appointments
Events	Google Sheet	Analytics

5.9 Live Inventory

Google Sheet Inventory tab (preferred) or UNAVAILABLE_PRODUCTS env. Toggle stock without redeploy.

5.10 Admin Dashboard

Tabs: Overview, Analytics, Sales, Leads, Appointments, Orders, Quotes, Inventory. Includes sync knowledge, test email, resume AI.

5.11 Other Features

- Email notifications (Resend) for handoff and delivery
- Reply-to message support (swipe YES maps to specific bot message)
- Smart welcome (Get Started + first message handling)
- Weekend/shop hours logic (Mon–Fri shop; agents daily 9 AM–9 PM)
- Payment proof flow (GCash / UnionBank)

6. Source of Truth — Where to Change What

Business facts → Google Doc

Edit: hours, prices, FAQ, products, payment, contacts, policies.

Workflow: Edit Google Doc → Save → Sync via /admin/sync-knowledge → No deploy needed.

Repo backup: knowledge/templates/beantol-knowledge-base.md (not auto-used in production).

Bot behavior → Code (git deploy)

Change	File(s)
Handoff, tone, sales script	system-rules.js
Delivery zone logic	lib/cebu-area-delivery.js, lib/outside-cebu-delivery.js
Welcome messages	lib/welcome.js
Quotes	lib/quotes.js, lib/post-quote-flow.js
Appointments	lib/appointments.js
Recommendations	lib/recommendations.js
Main pipeline	server.js

Stock → Google Sheet Inventory or Render env

Customer records → Google Sheets (not AI knowledge)

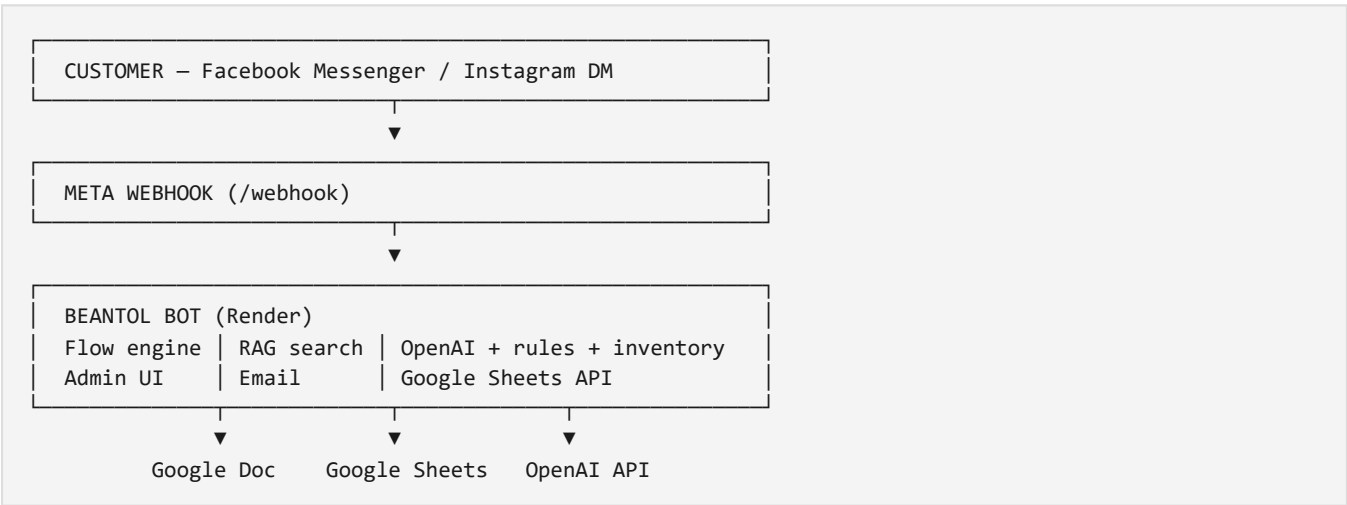
Quick guide

You want to...	Do this
Add FAQ / change price	Google Doc → sync
Fixed reply for delivery zone	Code → deploy
Mark bean out of stock	Inventory Sheet
Change appointment steps	Code → deploy

7. Technology Stack

Layer	Technology
Runtime	Node.js 18+
Web server	Express
AI	OpenAI API (gpt-4o-mini)
Embeddings	text-embedding-3-small (RAG)
Messaging	Meta Graph API
Knowledge	Google Drive API (Docs export)
CRM data	Google Sheets
Email	Resend (production)
Hosting	Render

8. Architecture



9. Knowledge Base (RAG)

1. Google Doc indexed into searchable chunks (embeddings)
2. Each message finds the 6 most relevant chunks
3. Chunks + behavior rules sent to OpenAI
4. AI writes a natural reply grounded in your content

Team edits Google Doc without code changes. Sync after edits.

10. Human Handoff Flow

Customer asks for human → handoff message + email
→ AI still answers follow-ups
→ Staff replies in Business Suite → bot PAUSES
→ After idle or Resume AI → "assistant is back" message

11. Costs & Infrastructure (Approximate)

Service	Notes
Render	Free tier or ~\$7+/mo always-on
OpenAI	Per-message; gpt-4o-mini is low cost
Resend	Free tier for low email volume
Google Cloud	Docs + Sheets API free tier usually enough
Meta	No per-message fee

12. What's Built vs Future

Done

AI + RAG, Messenger + Instagram, handoff, delivery zones, quotes/orders/leads/appointments, live inventory, admin dashboard, analytics, recommendations wizard, reply-to context, conversation memory.

Not started

WhatsApp, website widget, full CRM, business intelligence, visual workflow builder.

13. Maintenance Checklist

- **Weekly:** Check handoffs, stale leads, inventory
- **Business info changes:** Google Doc → sync → test
- **Behavior changes:** Code → deploy → test flow

14. Pitch for Other Shops

Any product-based business with repeated DMs (retail, F&B, roastery, boutique) benefits from:

- 24/7 professional first response
- Consistent pricing and policy answers
- Structured order/inquiry capture
- Human takeover when it matters
- Team-editable knowledge (Google Doc)
- Admin dashboard without building a full app

To build one: Meta Developer app, OpenAI account, cloud host, Google Workspace (optional), custom development for products and rules.

15. Admin URLs & Security

Purpose	URL pattern
Admin dashboard	/admin?token=ADMIN_SECRET
Sync knowledge	/admin/sync-knowledge?token=ADMIN_SECRET
Knowledge status	/admin/knowledge-status?token=ADMIN_SECRET
Test email	/admin/test-email?token=ADMIN_SECRET

Security: Never commit .env; keep ADMIN_SECRET private; rotate tokens if exposed.
